Binomial expansion is taking a two-term polynomial (binomial) and multiplying it by itself or raising it to a power.

$$(x+1)^2 = (x+1)(x+1)$$

$$(x+1)^3 = (x^2+2x+1)(x+1)$$

$$(x+1)^4 = (x^3+3x^2+3x+1)(x+1)$$

$$= x^4+4x^3+6x^2+4x^1+1$$

Multiplication distributes over addition

$$2(a+b) = 2a + 2b$$

$$(x \cdot y)^2 = x^2 \cdot y^2$$

Exponentiation distributes over multiplication and NOT over addition.

Exponentiation does not distribute over addition.

$$(x+n)^2 \neq x^2+n^2$$

$$(1+2)^2 \neq 1^2+2^2$$

$$9 \neq 5$$

For exponentiation over addition, you have to multiply it out, distributing each term over each term.

For binomials, each term has to be multiplied by each term.

$$(x+n)^2 = (x+n)(x+n)$$

$$= x^2 + 2nx + n^2$$

$$(1+2)^2 = (1+2)(1+2)$$

$$3^2 = 1\cdot 1 + 1\cdot 2 + 2\cdot 1 + 2\cdot 2$$

$$9 = 1 + 2 + 2 + 4$$

We do multiplication of each term by each term.

$$(x+n)^2$$

$$(x+n)(x+n)$$

$$x^2+2nx+n^2$$

$$* \qquad x+n$$

$$x^3+2nx^2+xn^2$$

$$+ \quad nx^2+2n^2x+n^3$$

$$x^3+3nx^2+3n^2x+n^3$$

From $(x+n)^2$ to $(x+n)^3$ notice that you go across left to right increasing by one power of x with the original coefficients. Then go across from left to right increasing by one power of n with the original coefficients. Then when you add like terms, you get new coefficients in a pattern.

$$
\begin{array}{r}
* \quad x^2 + 2nx + n^2 \\
x + n \\
\hline
x^3 + 2nx^2 + n^2x \\
+ \quad nx^2 + 2n^2x + n^3 \\
\hline
x^3 + 3nx^2 + 3n^2x + n^3
\end{array}
$$

Binomial expansion is taking a two-term binomial and multiplying it by itself or raising it to a power.

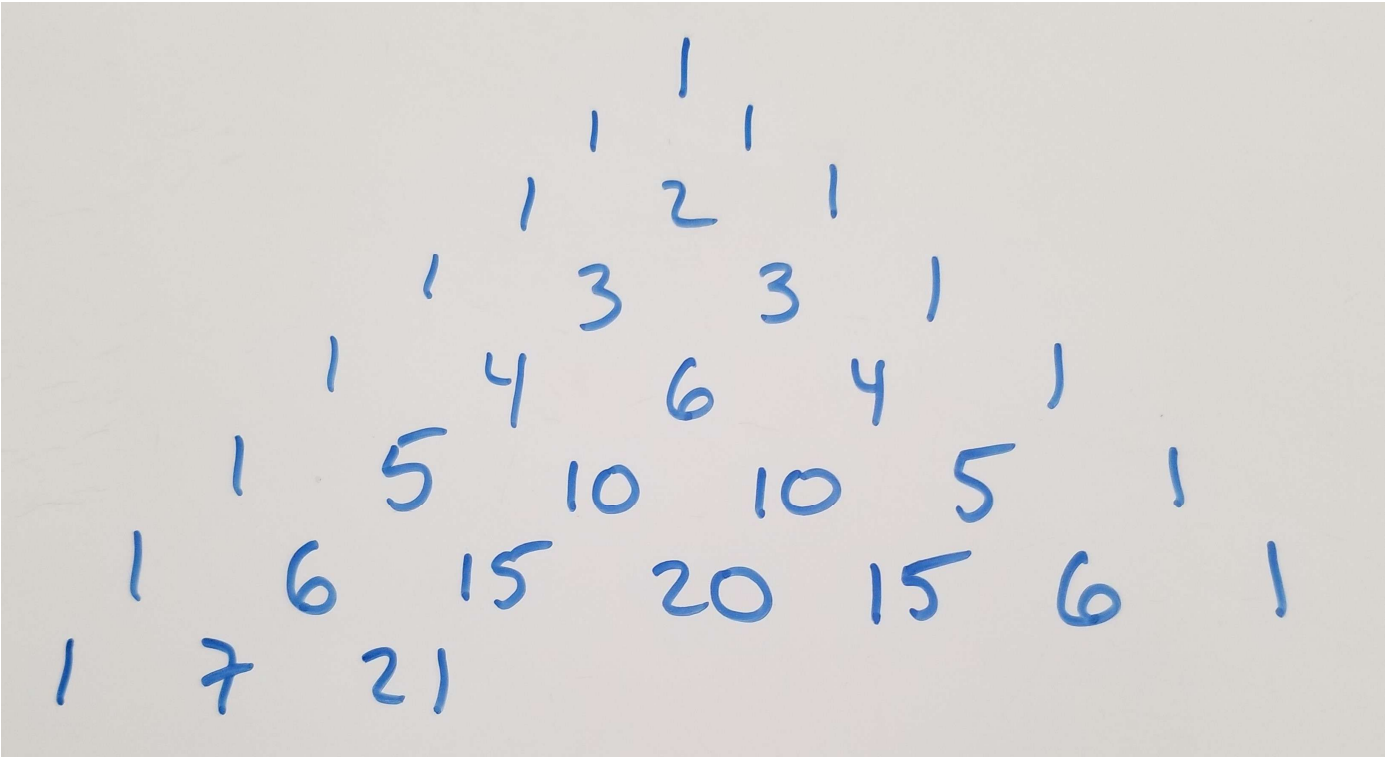
$$(x+1)^2 = (x+1)(x+1)$$

$$(x+1)^3 = (x^2+2x+1)(x+1)$$

$$(x+1)^4 = (x^3+3x^2+3x+1)(x+1)$$

$$= x^4+4x^3+6x^2+4x+1$$

# Pascal’s triangle is the pattern for the coefficients.



[Pascal's triangle – Wikipedia]

# Pascal's triangle

From Wikipedia, the free encyclopedia

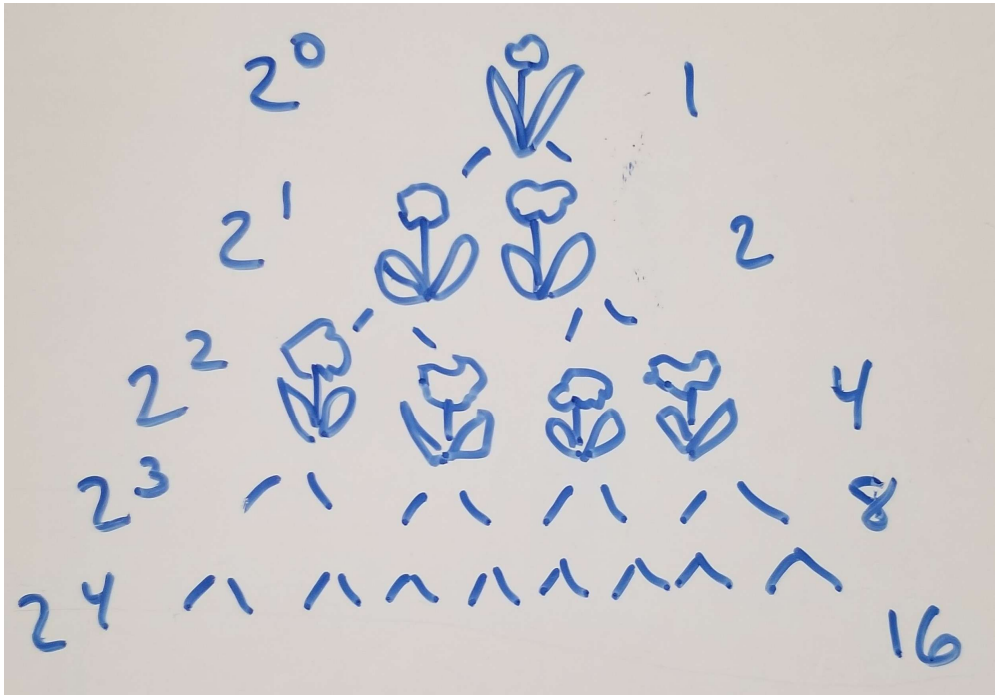
In mathematics, **Pascal's triangle** is a triangular array of the binomial coefficients that arises in probability theory, combinatorics, and algebra. In much of the Western world, it is named after the French mathematician Blaise Pascal, although other mathematicians studied it centuries before him in India,[1] Persia,[2] China, Germany, and Italy.[3]

The rows of Pascal's triangle are conventionally enumerated starting with row $n = 0$ at the top (the 0th row). The entries in each row are numbered from the left beginning with $k = 0$ and are usually staggered relative to the numbers in the adjacent rows. The triangle may be constructed in the following manner: In row 0 (the topmost row), there is a unique nonzero entry 1. Each entry of each subsequent row is constructed by adding the number above and to the left with the number above and to the right, treating blank entries as 0. For example, the initial number in the first (or any other) row is 1 (the sum of 0 and 1), whereas the numbers 1 and 3 in the third row are added to produce the number 4 in the fourth row.

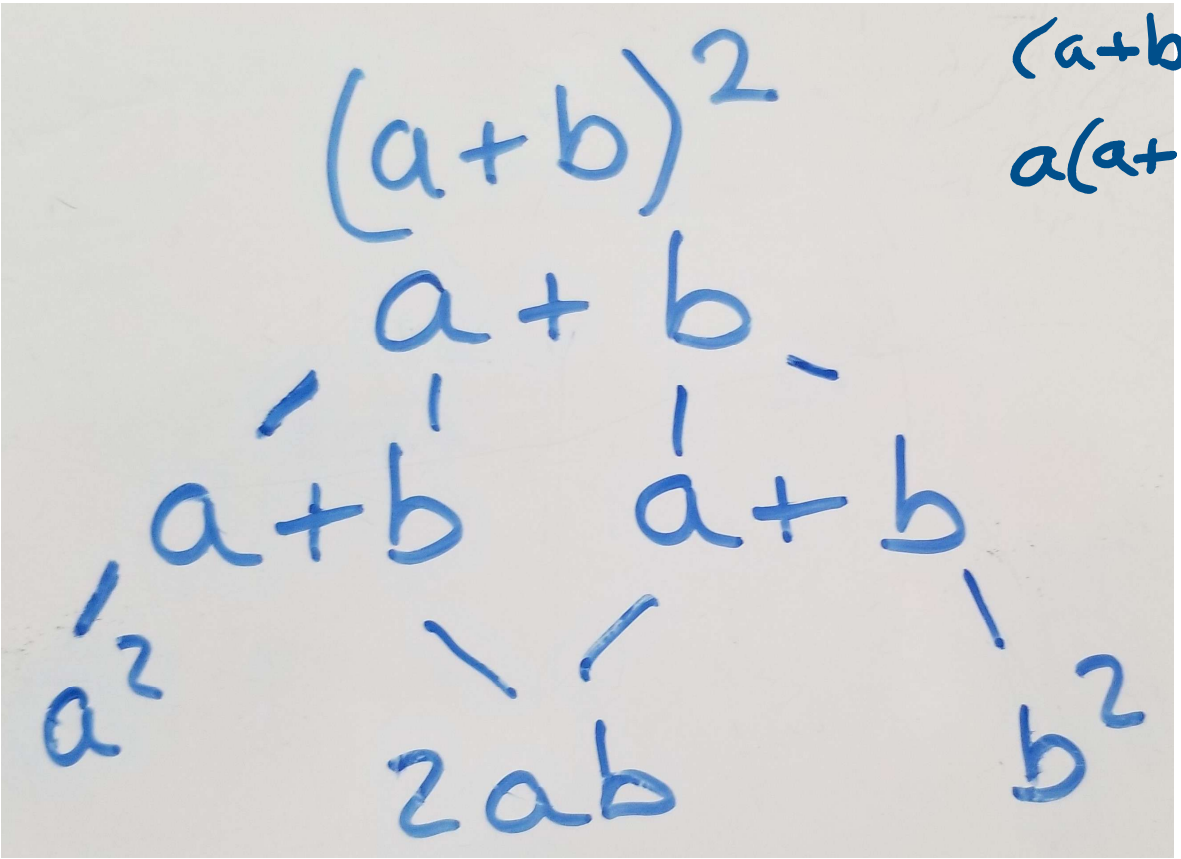
$$
\begin{array}{ccccccccccccccc}
 & & & & & & & 1 & & & & & & & \\
 & & & & & & 1 & & 1 & & & & & & \\
 & & & & & 1 & & 2 & & 1 & & & & & \\
 & & & & 1 & & 3 & & 3 & & 1 & & & & \\
 & & & 1 & & 4 & & 6 & & 4 & & 1 & & & \\
 & & 1 & & 5 & & 10 & & 10 & & 5 & & 1 & & \\
 & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 & \\
1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1
\end{array}
$$

A diagram showing the first eight rows of Pascal's triangle, numbered row 0 through row 7.

Exponents as generations: if you start with one plant and it creates two in the next generation, then you can draw a branching diagram with each level representing a generation, which is how many times the plant has doubled. These diagrams are often called trees, as they branch but they don’t look like trees since they are upside down.



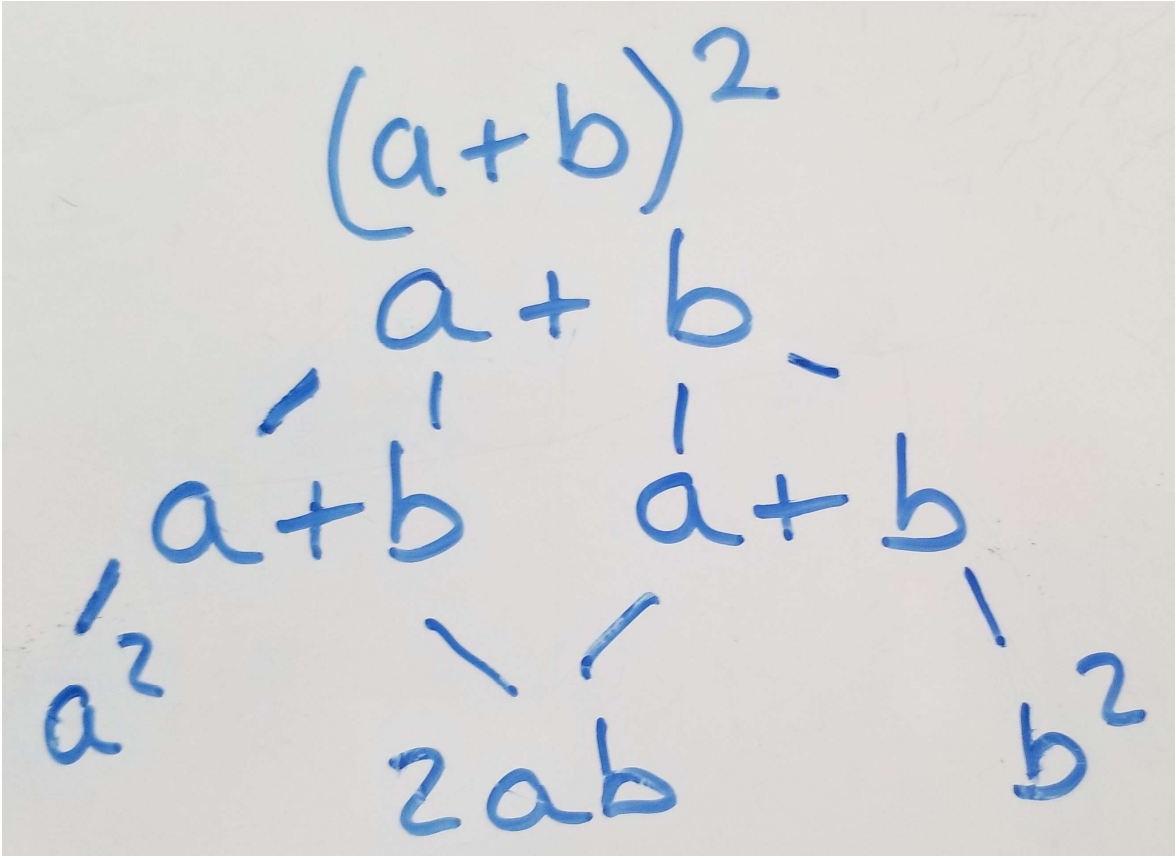
There is another way of thinking about this that makes it easier to see patterns and program a computer to do higher powered computations.
It uses a branch of maths called combinatorics.




$(a+b)(a+b)$

$a(a+b)+b(a+b)$

For $(a+b)^2$ thinking in terms of combinatorics, if there are 2 people and they can each have an apple and a banana, how many different combinations of apples and bananas can you get?

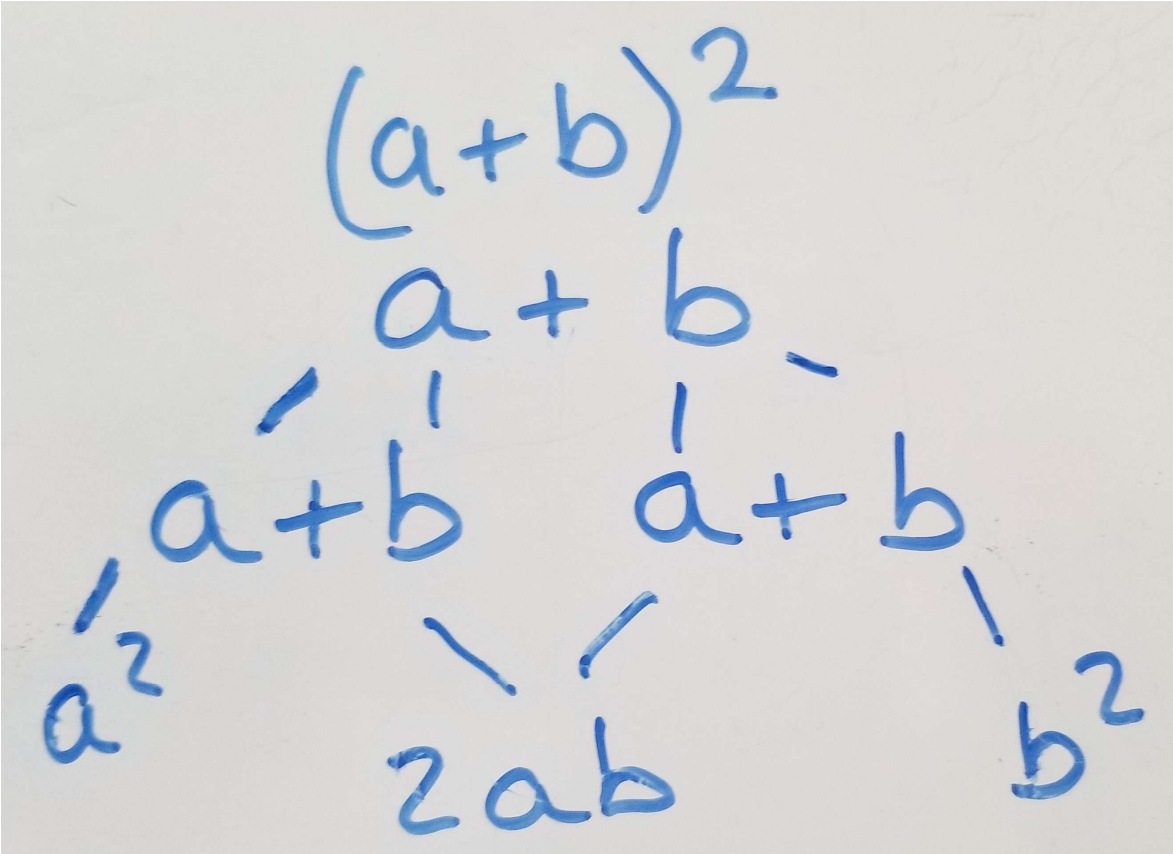


For two people each having one apple and one banana, if you want all combinations:

Both can have an apple,

Two ways of each person having one apple and one banana

Both can have a banana

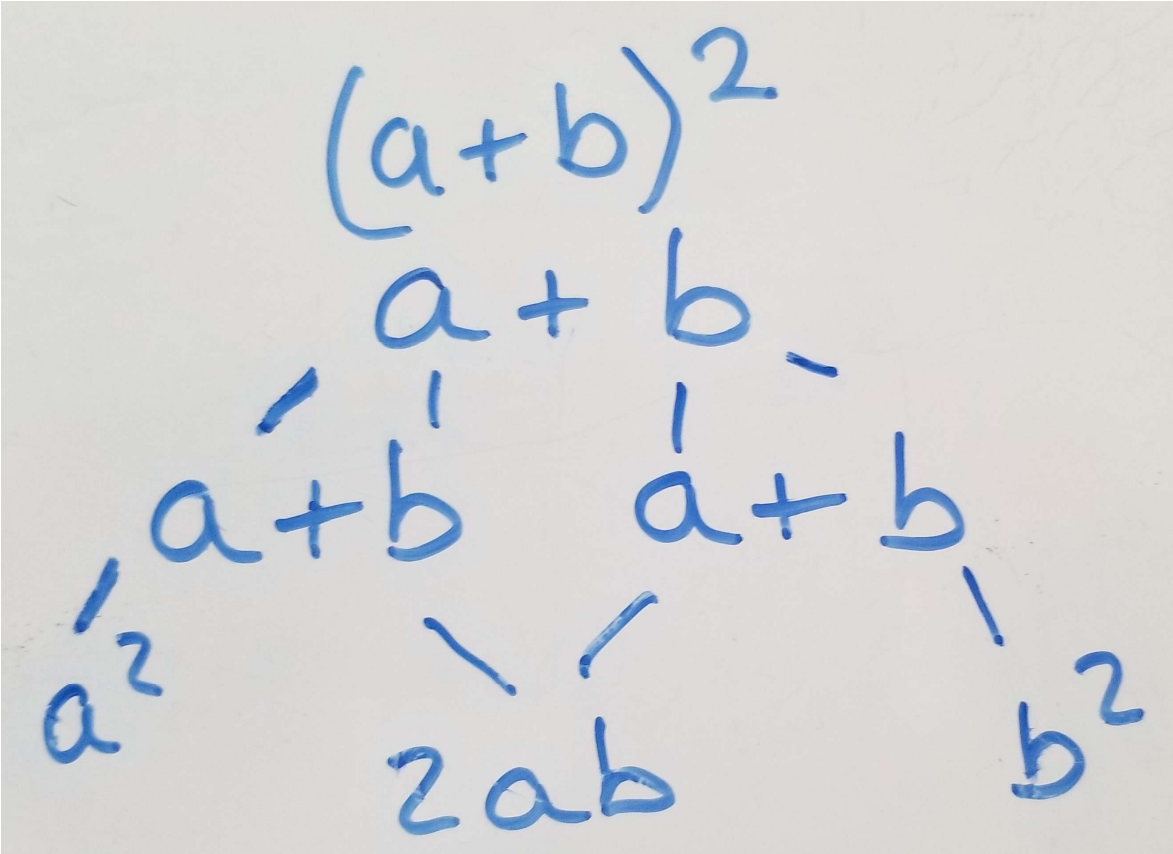


For two binomials and you want to multiply all the terms:

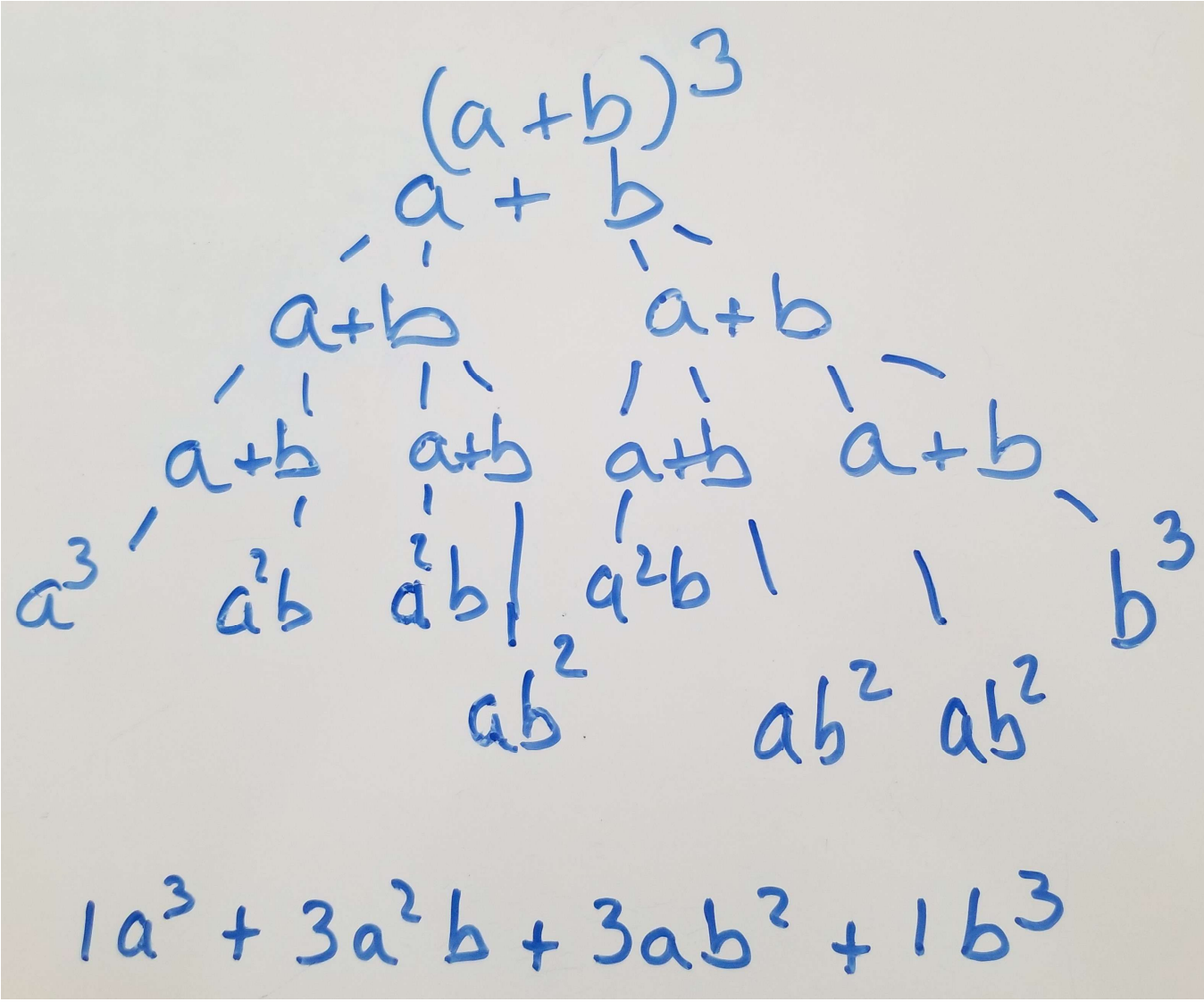
Can multiply a from each binomial

Two ways to get ab: ab or ba

Can multiply b from each binomial



For three binomials, you want to multiply all the terms:
Can multiply *a* from each binomial
Three ways to get *aab*: *aab, aba, baa*
Three ways to get *abb:* *abb, bab, bba*
Can multiply *b* from each binomial



There is a cool pattern that is based on Pascal's triangle, which comes from combinatorics.

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

1 7 21

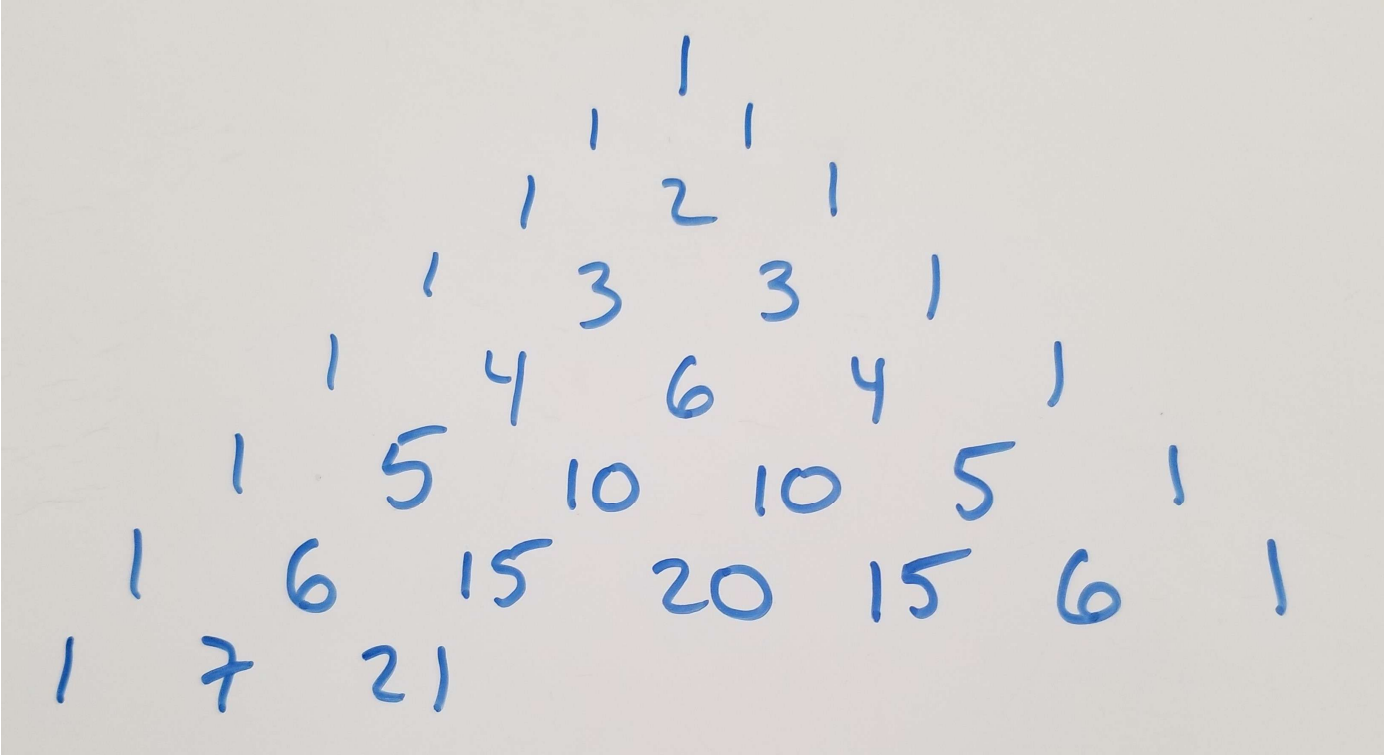
$$(x+n)^0 = 1$$

$$(x+n)^1 = x+n$$

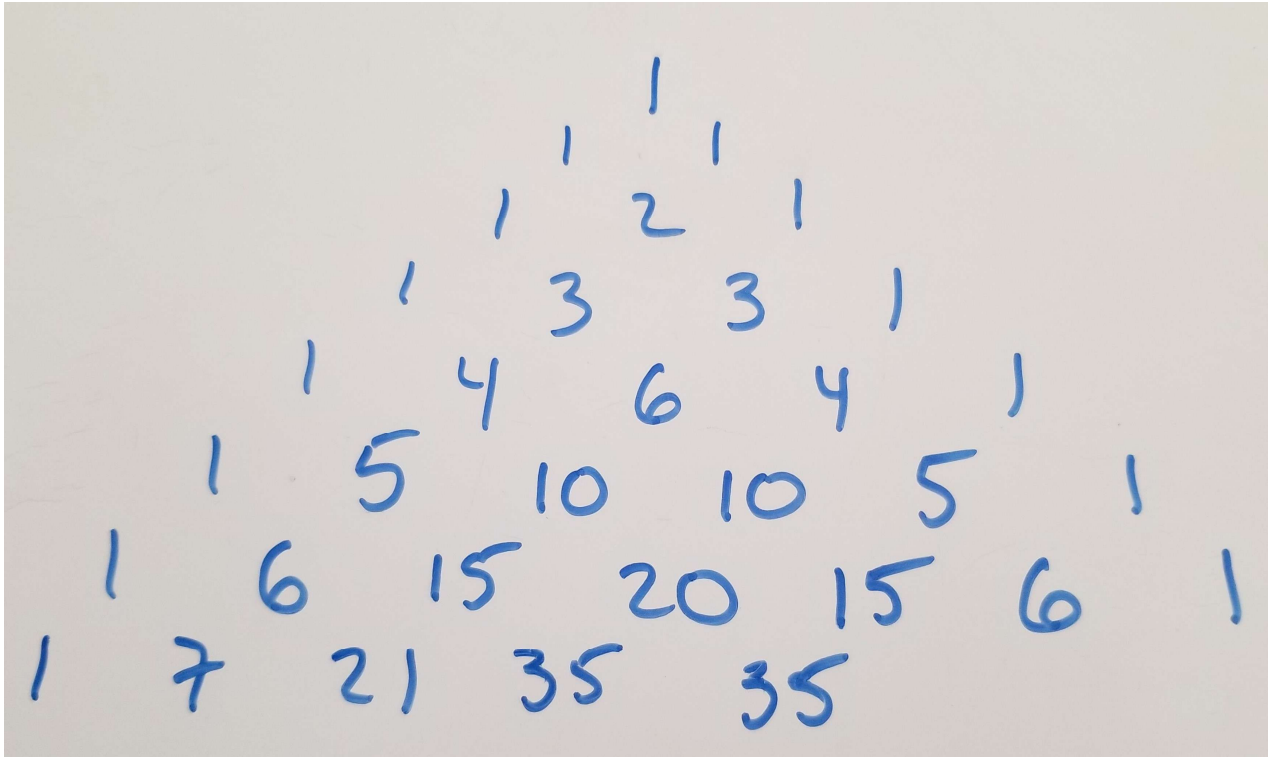
$$(x+n)^2 = x^2+2nx+n^2$$

$$(x+n)^3 = x^3+3nx^2+3n^2x+n^3$$

# Pascal's triangle – Wikipedia



# The coefficients of the binomial expansion can be gotten from Pascal's triangle.



Pascal's triangle gives the coefficients for the binomial expansion.

$$
\begin{array}{ccccccccccccc}
 & & & & & & 1 & & & & & & \\
 & & & & & 1 & & 1 & & & & & \\
 & & & & 1 & & 2 & & 1 & & & & \\
 & & & 1 & & 3 & & 3 & & 1 & & & \\
 & & 1 & & 4 & & 6 & & 4 & & 1 & & \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1 & \\
1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\
1 & 7 & & 21 & & 35 & & 35 & & & & &
\end{array}
$$

$$1$$

$$x+n$$

$$x^2+2nx+n^2$$

$$x^3+3nx^2+3n^2x+n^3$$

$$x^4+4nx^3+6n^2x^2+4n^3x+n^4$$

$$x^5+5nx^4+10n^2x^3+10n^3x^2+5n^4x+n^5$$

There is a pattern for the variable and the second part of the binomial.

$$1$$

$$x+n$$

$$x^2+2nx+n^2$$

$$x^3+3nx^2+3n^2x+n^3$$

$$x^4+4nx^3+6n^2x^2+4n^3x+n^4$$

$$x^5+5nx+10nx+10nx+5nx+n^5$$

The x power goes down, and the n power goes up with the coefficients from Pascal's triangle.

$$1$$

$$x+n$$

$$x^2+2nx+n^2$$

$$x^3+3nx^2+3n^2x+n^3$$

$$x^4+4nx^3+6n^2x^2+4n^3x+n^4$$

$$x^5+5n^1x^4+10n^2x^3+10n^3x^2+5n^4x+n^5$$

Notice the pattern of the second term for each power. This will be used when we learn to take the derivative.

$$(x+n)^p$$

$p=0$ $\quad 1$

$p=1$ $\quad x+n$

$p=2$ $\quad x^2+2nx+n^2$

$p=3$ $\quad x^3+3nx^2+3n^2x+n^3$

$p=4$ $\quad x^4+4nx^3+6n^2x^2+4n^3x+n^4$

$p=5$ $\quad x^5+5nx^4+10n^2x^3+10n^3x^2+5n^4x+n^5$

Notice the pattern of the second term for each power. This will be used when we learn to take the derivative.

$$(x+\Delta x)^p$$

$$p=0 \quad 1$$

$$p=1 \quad 1x + 1\Delta x$$

$$p=2 \quad 1x^2 + 2x\Delta x + \Delta x^2$$

$$p=3 \quad 1x^3 + 3x^2\Delta x + 3x\Delta x^2 + \Delta x^3$$

$$p=4 \quad 1x^4 + 4x^3\Delta x + 6x^2\Delta x^2 + 4x\Delta x^3 + \Delta x^4$$